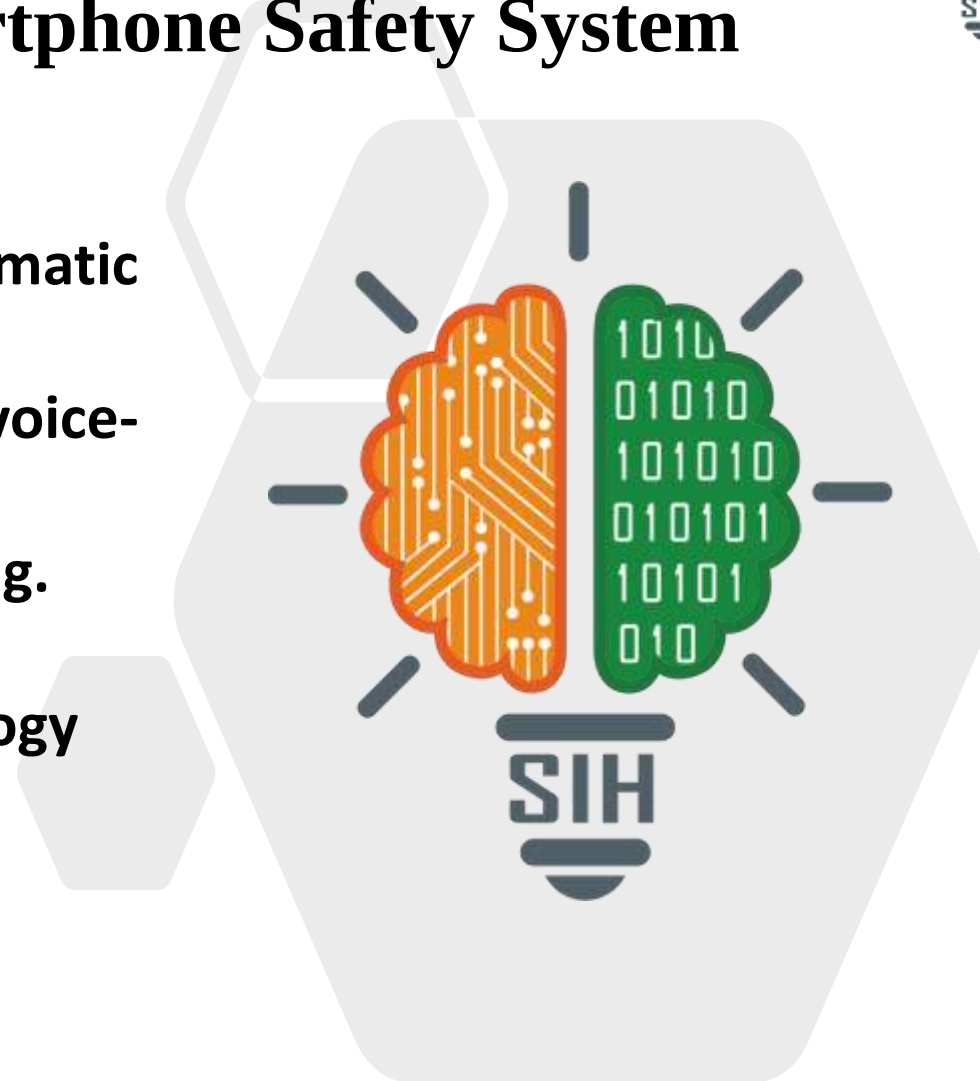


SMART INDIA HACKATHON 2026



INVERO | Smartphone Safety System

- Problem Statement ID – SIH26216
- Problem Statement Title – Automatic phone-snatching detection and voice-triggered SOS with live location sharing.
- Theme – Public safety using technology
- PS Category – Hardware
- Team ID – 165699
- Team Name – Invero



Smartphone Voice Assistance Safety System

Motion-based phone-snatching detection + personalized voice code-word SOS

Detailed explanation

- Detection logic compares motion with a snatch signature
- User sets an emergency code word during setup
- Saying the code word triggers SOS + live location sharing

How it addresses the problem

- Detects a snatch without requiring a button press
- Voice code word provides hands-free emergency activation
- SOS sends live location to predefined emergency contacts

Innovation and uniqueness

- Software-based motion-pattern detection
- Accelerometer + gyroscope cuts false alarms
- No extra hardware, low-cost, smartphone-native

Detection flow



Technologies to be used

Built-in phone sensors

- 3-axis accelerometer
- 3-axis gyroscope

Mobile app

- Safety-mode interface
- Monitoring + alert handling

Detection logic

- Motion processing
- Pattern check + ML classification
- Voice detection

Methodology and process



Sensor window → preprocessing → ML inference → confidence threshold → alert / continue monitoring

Feasibility

Feasible hardware

Phone's built-in gyroscope gives the motion signals; no add-on needed.

Feasible software

Mobile app + ML inference
Lightweight Random Forest classifier can run on-device and be refined through testing.

Scalable path

Roadmap from working prototype to field-ready app.

Challenges and risks

False detection
Voice recognition errors

Different snatching patterns
Snatch motion varies between incidents

Sensor noise
Raw signals can be noisy

Real-world reliability
Performance may drop outside test conditions

Strategies to overcome

Combine acceleration + gyroscope; user code word + confirmation

Train + validate ML model with varied snatch / non-snatch samples

Calibration + filtering

Repeated scenario testing

PROTOTYPE → TEST → CALIBRATE → IMPROVE → SCALE

Impact on target audience

- Faster detection and less dependence on manual SOS
- Hands-free voice code-word emergency activation
- Live location shared with predefined emergency contacts

Social

- Faster response in sudden incidents
- Greater personal-safety confidence

Economic

- No extra hardware cost for users
- Fewer losses from stolen phones

Environmental

- Uses the smartphone people already carry
- No add-on hardware, so no extra e-waste

KEY IMPACT • Reduced response delay during a sudden incident

01

Android motion sensors (gyroscope) guide

[Android Developers: developer.android.com/develop/sensors-and-location/sensors/sensors_motion](https://developer.android.com/develop/sensors-and-location/sensors/sensors_motion)

02

Sensor API documentation

[Android SensorManager reference: developer.android.com/reference/android/hardware/SensorManager](https://developer.android.com/reference/android/hardware/SensorManager)

03

Activity recognition research

Accelerometer + gyroscope based movement classification literature

04

SIH 2026 framework

[Official Smart India Hackathon guidelines: sih.gov.in](https://sih.gov.in)

Research focus: calibration • filtering • motion-pattern analysis • ML classification • false-detection reduction